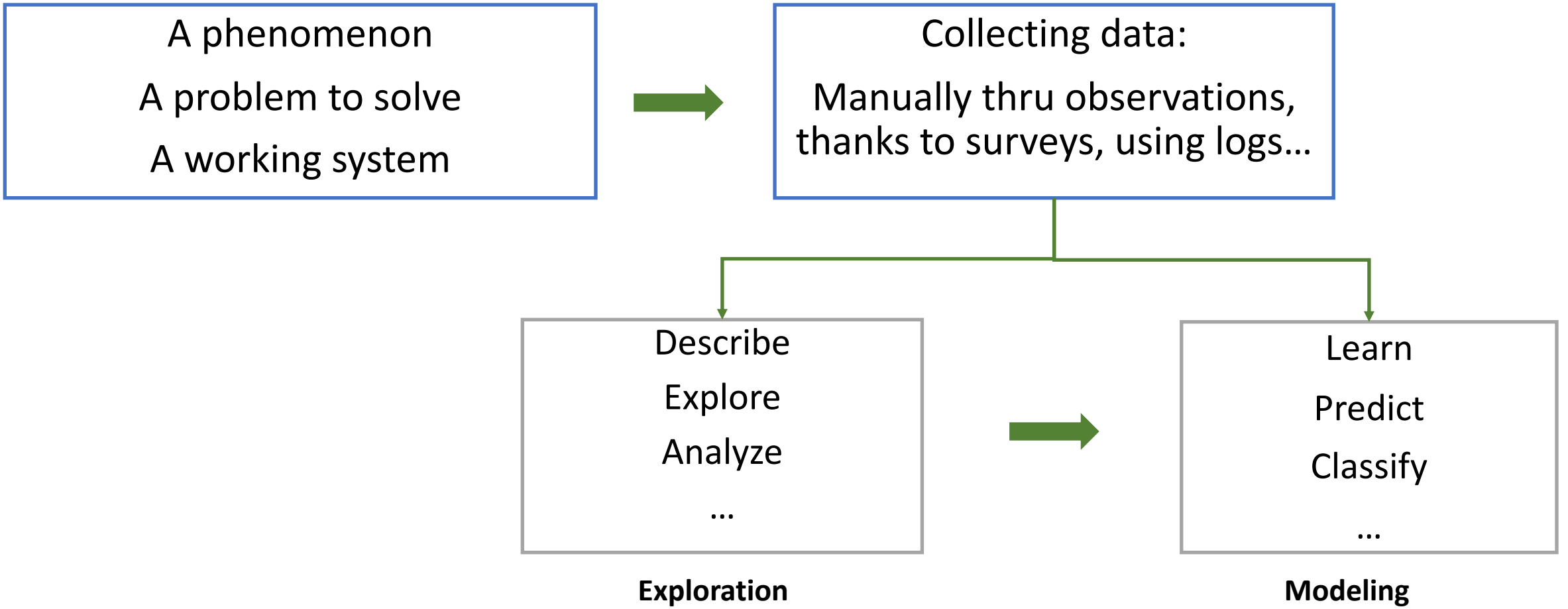
Course
« *Deep Learning*»

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ENSIA

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Introduction



A brief introduction to data exploration and analysis

1. *High dimensional data*
2. *Principal component analysis*
3. *Unsupervised classification (clustering)*

Introduction

- Descriptive analysis:
  - Exploration
  - Data description
  - Focus on a sample
- We do not assume any prior model.
- As soon as we would like to generalize the analysis to all the data:

→ Modeling

Generalization  Modeling

Introduction

- **Example:**

  Data about cars.

Description:

A automobile rental company would like to know the condition of its fleet.

- Diesel cars have a higher mileage than gasoline cars.

Modeling:

Used car sales website.

- Estimate the price of a car from its characteristics.

Introduction

- **Example:**

  Covid-19: Data of a set of people is available

Description:

Analyze, describe, explore…

People with chronic diseases (such as diabetes or HTA) develop more acute symptoms.

Introduction

- **Example:**

  Covid-19: Data of a set of people is available

Modeling :

Prediction, classification… towards generalization.

If you consider, age, socio-professional category, family situation...
What is the probability of being infected with Covid-19?

Introduction

- Descriptive analysis concerns a set of individuals: A population.
- Individuals are not necessarily people:
  - objects;
  - animals;
  - images;
  - videos;
  - sensor surveys;
  - …
- And we do not study the population as a whole.
  - You can « observe » everything.
  - The study most often focuses on a sample.

Introduction

- Studying an individual/object => analyze its features/properties.
  - One feature → a variable

- The sample to be studied is represented as a table:
  - Rows = individuals
  - Columns = variables.

Sujet	Math	Sciences	Français	Latin	Musique
Jean	6	6	5	5,5	8
Aline	8	8	8	8	9
Annie	6	7	11	9,5	11
Monique	14,5	14,5	15,5	15	8
Didier	14	14	12	12	10
André	11	10	5,5	7	13
Pierre	5,5	7	14	11,5	10
Brigitte	13	12,5	8,5	9,5	12
Evelyne	9	9,5	12,5	12	18

1. High-dimensional data

- One-dimensional analysis

  => Study variables one by one, and separately.

- With two variables

  => we can see, for example, the correlation.

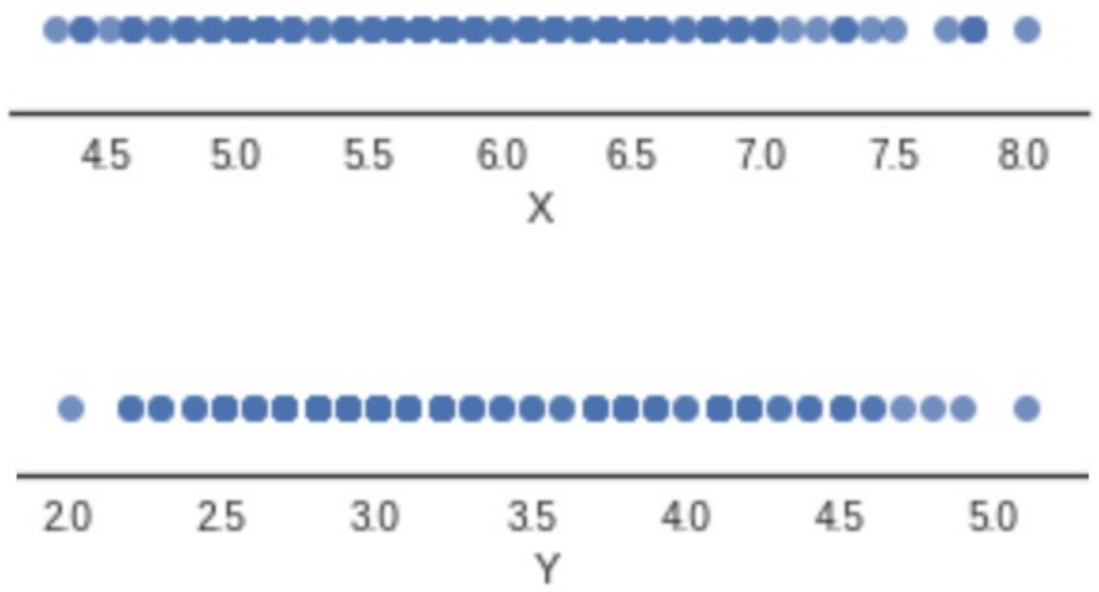
- More variables analyzed jointly: multidimensional analysis.

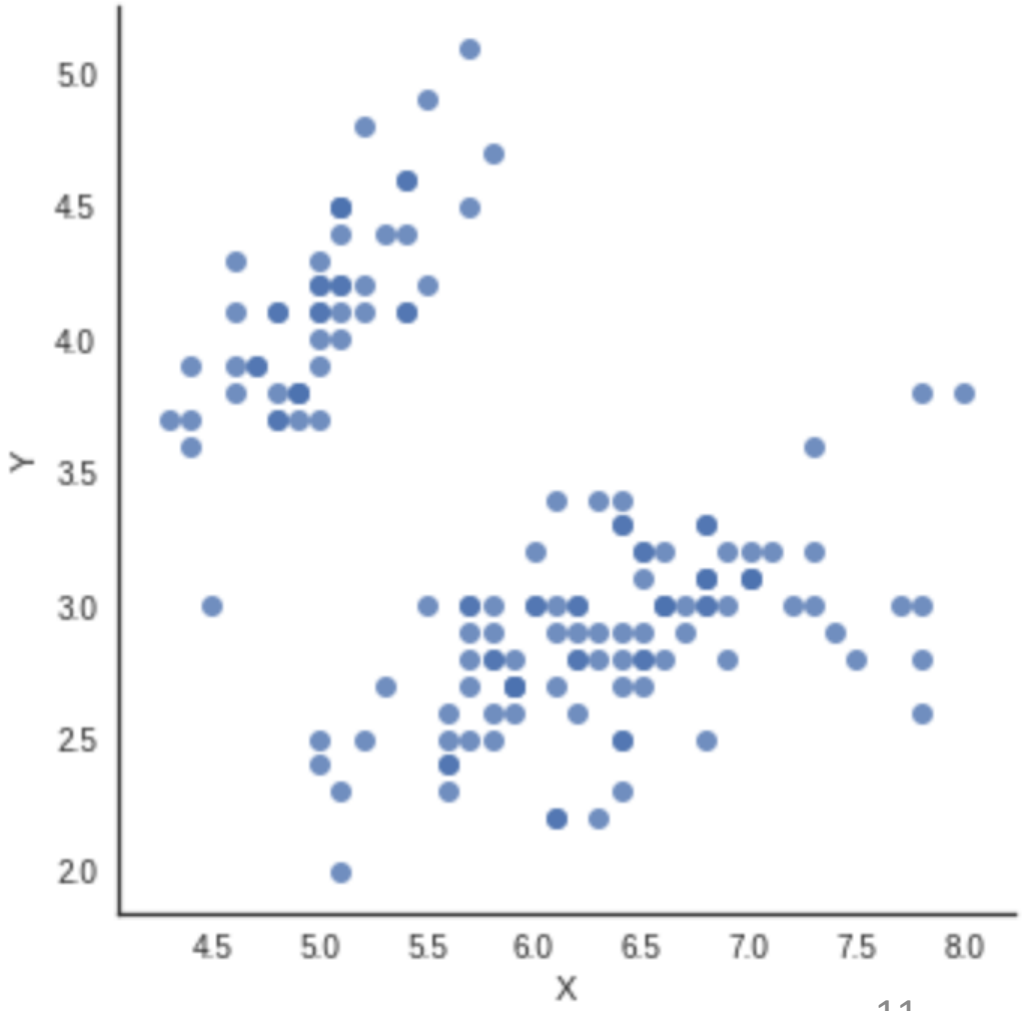
  => Allows you to see more things!

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1. High-dimensional data

- Individuals: *I (X,Y)*





1. High-dimensional data

- Dimension > 3: Brain unable to perceive and represent data.

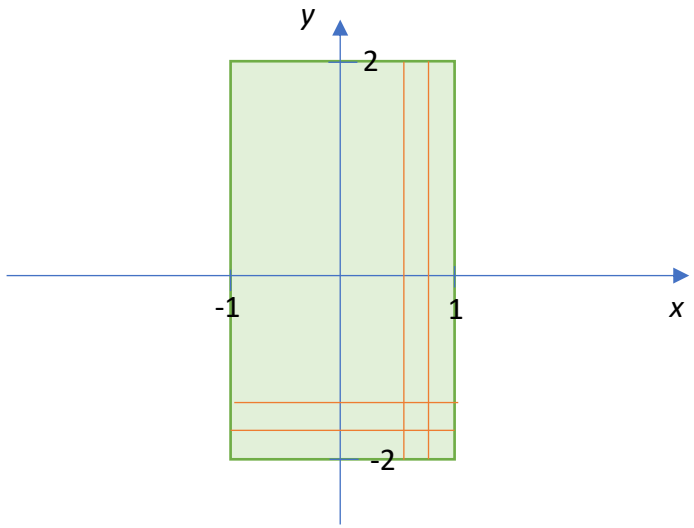
- High-dimensional data has properties that are beyond our intuition.

→ ***Dimensionality Curse.***

1. High-dimensional data

Dimensionality curse

- *This refers to the difficulties that arise in high-dimensions when it comes to:*
  - *Analyze,*
  - *Explore,*
  - *Make use,*
  - *Manage,*
  - *Process,*
  - *…*



- Introduced for the first time by Richard Bellman [1961]:

  It refers to the difficulty to perform optimization in high-dimensional spaces using exhaustive methods.

1. High-dimensional data

Dimensionality curse

- Also used in statistics:

  It refers to the relationship between the size of the data sample and the reliability of the estimate [Donoho 2000].

- It has been shown that the number of samples must increase *exponentially* with the dimension in order to maintain the same level of reliability of the estimate.

1. High-dimensional data

Dimensionality curse

- When the dimension increases:
  - The distance to the nearest neighbor increases
  - The distance to the furthest neighbor increases
  - *d(nearest neighbor)* increases faster than *d(furthest neighbor)*

- Under certain conditions (including data uniformly distributed on all dimensions and independent):
  - When $d \to \infty$:

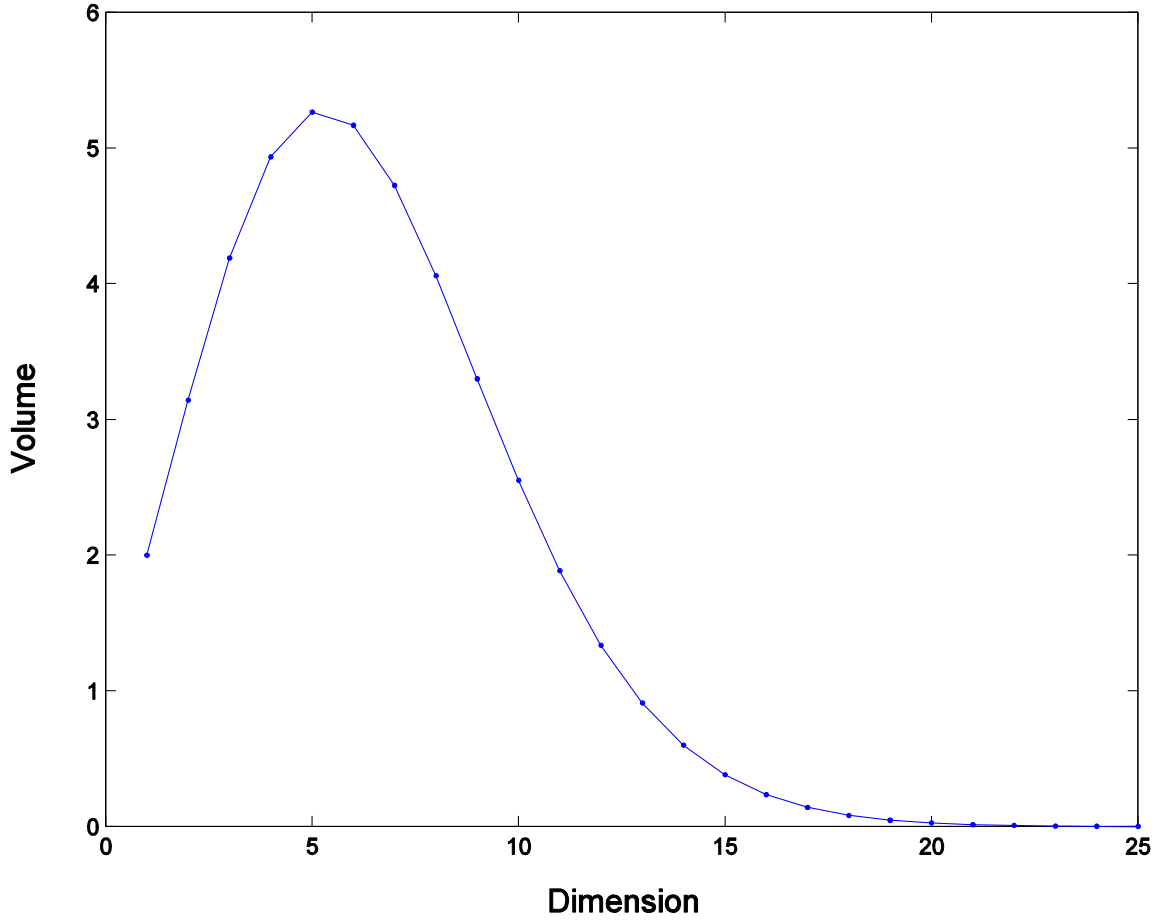
    Distance to the nearest neighbor = distance to the furthest neighbor!

    [Beyer 1999]

1. High-dimensional data

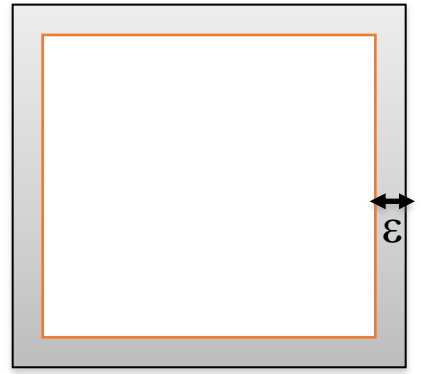
Dimensionality curse

- The volume of a hyper-sphere



1. High-dimensional data

Dimensionality curse



Exercise

Let’s consider a set of vectors uniformly distributed in a "hyper-square" (a 2D square, a 3D cube...).

Let’s consider the space inside the hyper-square at a distance ε from the edge: grey space (in the figure).

- If we draw a random point inside the hyper-square, what is the probability that the point falls in the gray space, if we consider that the space is d-dimensional?
- Study the variation of this probability according to the dimension: 2, 3, 4,..., 100. Draw the corresponding curve.
- What do you notice?

1. High-dimensional data

A summary

- Number of variables (dimension) +++
- Number of individuals +++

=> The analysis becomes difficult.

=> Less or no intuition

=> Costly processing

Two possible approaches:

1. Reduce the number of variable: Create new variables from the original ones while losing as little information as possible. **PCA**
2. Partition the data set (individuals) by grouping similar individuals ***Clustering***